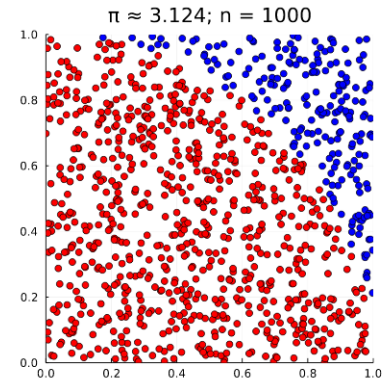


Anatomy of a Random Variable

A more in-depth treatment of many of these concepts can be found at: https://github.com/zsunberg/CU-DMUPP-Materials/blob/main/notes/STAT_219_Notes.pdf

Outline

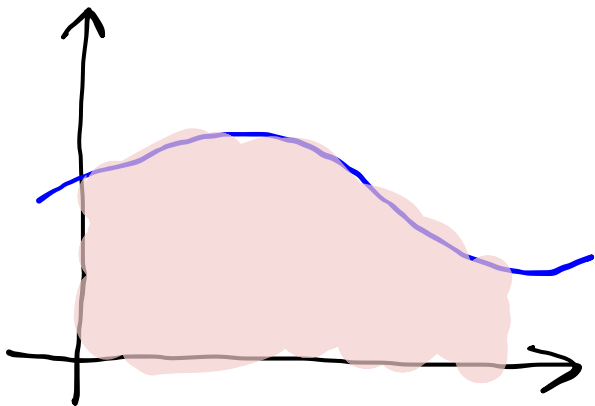
- A Motivating Example: Monte Carlo Integration
- Rigorous Definitions of a Random Variable
- Law of large numbers and the Central Limit Theorem



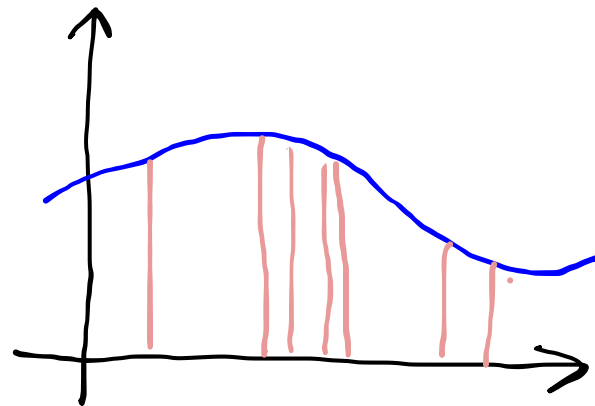
$$X : \Omega \rightarrow E$$

$$\sqrt{n} (\bar{X}_n - \mu) \xrightarrow{d} \mathcal{N}(0, \sigma^2)$$

Monte Carlo Integration



$$I = \int_{\Omega} f(x) dx$$
$$I = \int_{\Omega} f(x) \mu(dx)$$



$$X_i \sim U(\Omega)$$

$$I \approx Q_N \equiv \frac{\int_{\Omega} \mu(dx)}{N} \sum_{i=1}^N f(X_i)$$

A useful function: the Indicator

Monte Carlo Integration

Special Case: Expectation

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x p(x) dx \\ &\approx \frac{1}{N} \sum_{i=1}^N X_i \end{aligned}$$

In what sense does this converge as $N \rightarrow \infty$? How accurate is it?

Random Variables

Why are probability distributions not enough?

~~Consider this definition: *Two random variables are equal if their probability distributions are the same.*~~

$$A \sim \text{Bernoulli}(0.5)$$

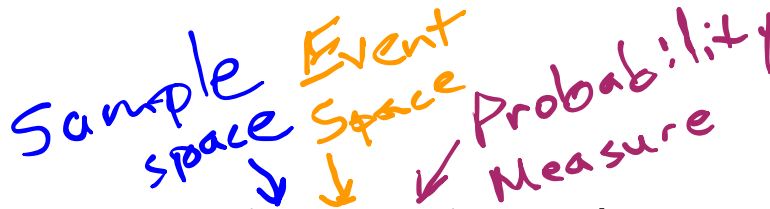
$$B = \neg A$$

$$B \stackrel{?}{=} A$$



Random Variables

Sample space Event Space Probability Measure



Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E -valued **random variable** is a **measurable function** $X : \Omega \rightarrow E$.

$$\omega \in \Omega$$

$$X(\omega) \in E$$

What is this function?

What is $\omega \in \Omega$?

Example: Coin World

$$\Omega = \{H, T\}$$

$$E = [0, 1]$$

$$X(\omega) = \mathbf{1}_{\{H\}}(\omega)$$



Example: Many coins

$$\Omega = \{H, T\}^\infty$$

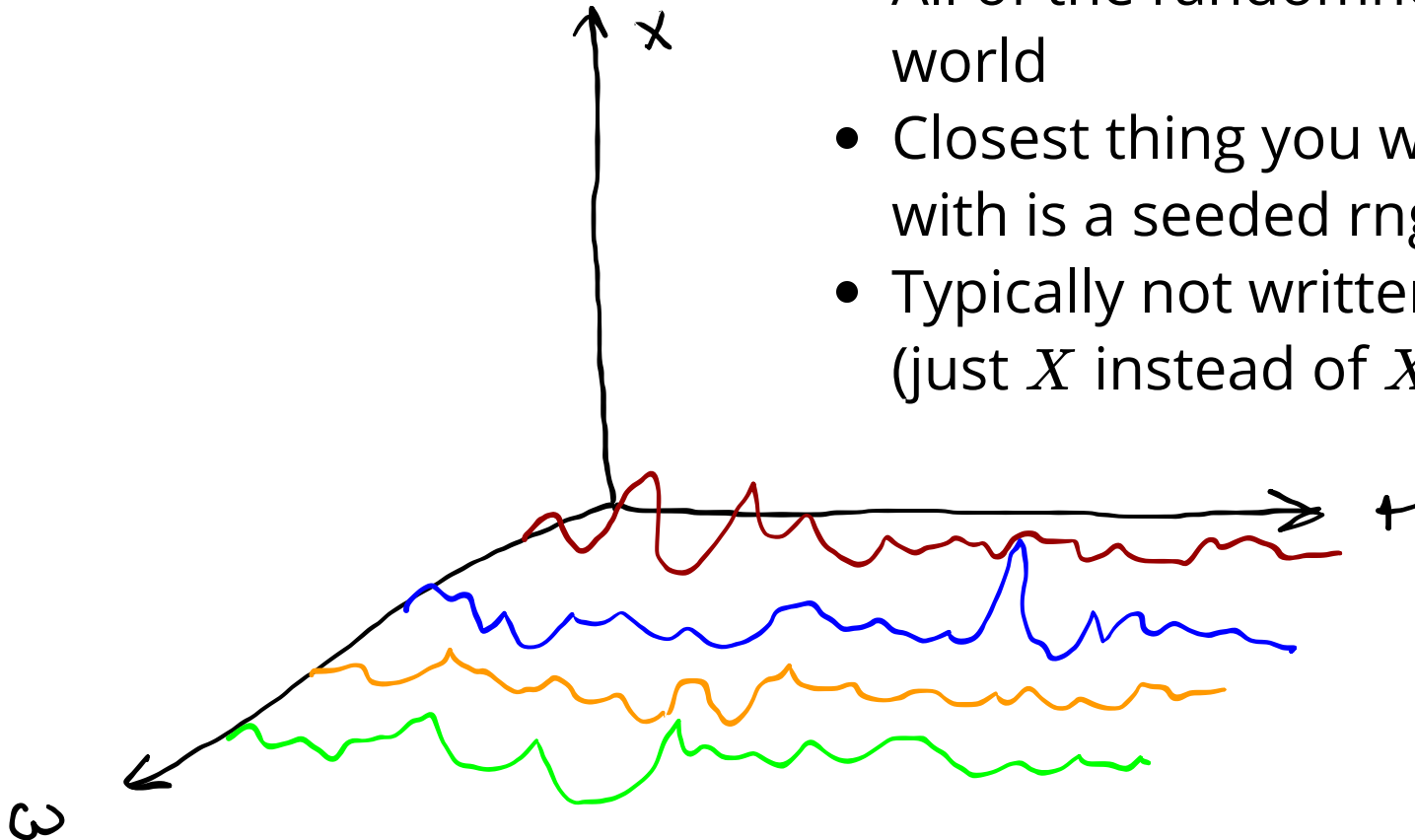
$$E = [0, 1]$$

$$X_i(\omega) = \mathbf{1}_{\{H\}}(\omega_i)$$



What is $\omega \in \Omega$?

- All of the randomness in the world
- Closest thing you work regularly with is a seeded rng
- Typically not written expressly (just X instead of $X(\omega)$)



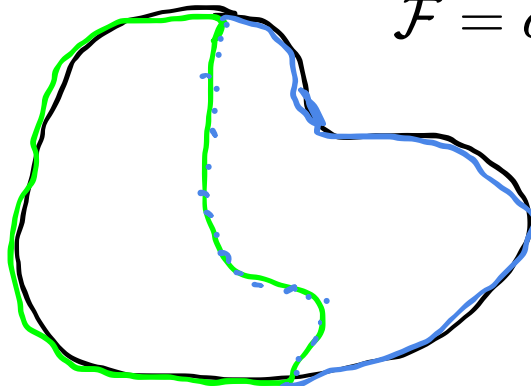
What is $\mathcal{F}$ (and $\mathcal{E}$)?

- "σ-algebra" or "σ-field"
- Subset of subsets of Ω (that is, $\mathcal{F} \subseteq 2^\Omega$)
- Three requirements to be a σ-field
 - $\Omega \in \mathcal{F}$
 - If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ (where $A^c = \Omega \setminus A$)
 - If $A_i \in \mathcal{F}$ for $i \in \mathbb{N}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$
- $\sigma(\cdot)$ creates a σ-field from a set of generators

$$\Omega: \{1, 2, 3\}$$

$$\mathcal{F}^\Omega = \{\{1\}, \{2\}, \{3\}, \emptyset, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Ω



$$\mathcal{F} = \sigma(\{A\}) = \{\Omega, A, A^c, \emptyset\}$$

σ -algebra Examples

If $\Omega = \{1, 2, 3\}$, what is $\sigma(\{1\})$?

If $\Omega = \{1, 2, 3\}$, what is $\sigma(\{1, 2\})$?

Borel Sigma Algebra

The Borel σ -algebra for a topological space Ω is the σ -field generated by all open sets in Ω .

The Borel σ -field on $\mathbb{R}$ is $\mathcal{B} = \sigma(\{(a, b) : a, b \in \mathbb{R}\})$

- Is $(1, 2)$ in $\mathcal{B}$?
- Is $(1, \infty)$ in $\mathcal{B}$?
- Is $[1, 2]$ in $\mathcal{B}$?
- Is 1 in $\mathcal{B}$?
- $\Omega \in \mathcal{F}$
- If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$ (where $A^c = \Omega \setminus A$)
- If $A_i \in \mathcal{F}$ for $i \in \mathbb{N}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

What is P ?

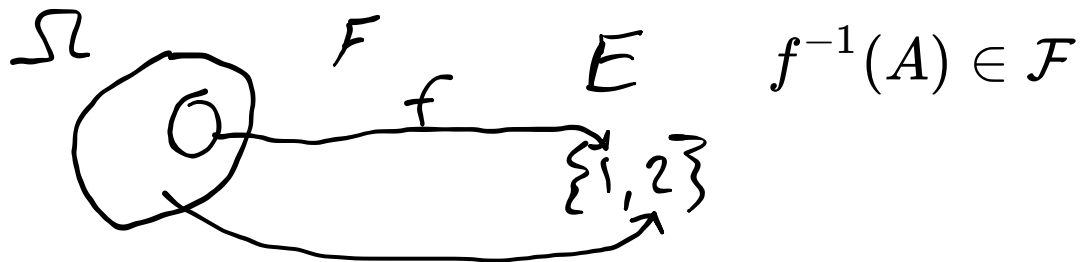
A probability measure P is a function $P : \mathcal{F} \rightarrow [0, 1]$ having the following properties:

1. $0 \leq P(A) \leq 1 \quad \forall A \in \mathcal{F}.$
2. $P(\Omega) = 1.$
3. (Countable additivity) $P(A) = \sum_{n=1}^{\infty} P(A_n)$ whenever $A = \cup_{n=1}^{\infty} A_n$ is a countable union of disjoint sets $A_n \in \mathcal{F}$

Random Variables

Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E -valued **random variable** is a **measurable function** $X : \Omega \rightarrow E$.

A function $f : \Omega \rightarrow E$ is measurable if for every $A \in \mathcal{E}$, the pre-image of A under f is in $\mathcal{F}$. That is, for all $A \in \mathcal{E}$

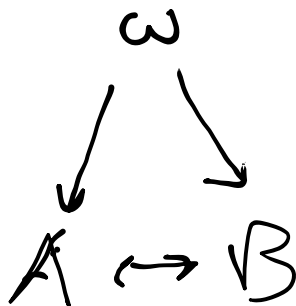


Are there functions that are not Borel-measurable?

Yes! Example: $\mathbf{1}_V$ where V is a [Vitali Set](#)

Advantages over pdf definition

- More sophisticated convergence concepts
- Better way of thinking about related random variables (personally, I think)



Break

Exercise 1.2.5. *Let $\Omega = \{1, 2, 3\}$. Find a σ -field $\mathcal{F}$ such that $(\Omega, \mathcal{F})$ is a measurable space, and a mapping X from Ω to $\mathbb{R}$, such that X is not a random variable on $(\Omega, \mathcal{F})$.*

A function $f : \Omega \rightarrow E$ is measurable if for every $A \in \mathcal{E}$, the pre-image of A under f is in $\mathcal{F}$. That is, for all $A \in \mathcal{E}$

$$f^{-1}(A) \in \mathcal{F}$$

Hint:

1. Use $\sigma(\{1\})$ (what is it?)
2. What function has a preimage not in $\sigma(\{1\})$?

$$\mathcal{F} = \{\Omega, \emptyset, \{1\}, \{2, 3\}\}$$

$$X = \mathbf{1}_{\{1,2\}}$$

Convergence

Review: For a (deterministic) sequence $\{x_n\}$, we say

$$\lim_{n \rightarrow \infty} x_n = x$$

or

$$x_n \rightarrow x$$

if, for every $\epsilon > 0$, there exists an N such that $|x_n - x| < \epsilon$ for all $n > N$.

Convergence

In what senses can we talk about a sequence of random variables converging?

- Sure ("pointwise")
- Almost Sure
- In Probability
- Weak ("in distribution"/"in law")

When are two R.V.'s the same?

$$X = Y \text{ if } X(\omega) = Y(\omega) \quad \forall \omega \in \Omega$$

In practice, there are often unimportant ω where this is not true.

We say that X is *almost surely* the same as Y if
$$P(\{\omega : X(\omega) \neq Y(\omega)\}) = 0.$$

This is denoted $X \stackrel{a.s.}{=} Y$ and the terms *almost everywhere* (a.e.) and *with probability 1* (w.p.1) mean the same thing.

Sure Convergence

$$X_n(\omega) \rightarrow X(\omega) \quad \forall \omega \in \Omega$$

Almost Sure Convergence

$X_n \xrightarrow{a.s.} X$ if there exists $A \in \mathcal{F}$ with $P(A) = 1$ such that
 $X_n(\omega) \rightarrow X(\omega)$ for each fixed $\omega \in A$.

Does sure convergence imply almost sure convergence?

Convergence in Probability

$X_n \rightarrow_p X$ if $P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \rightarrow 0$ for any fixed $\epsilon > 0$.

Does $X_n \xrightarrow{a.s.} X$ imply $X_n \rightarrow_p X$?

Yes.

Convergence in Probability

Does $X_n \rightarrow_p X$ imply $X_n \xrightarrow{a.s.} X$?

No.

PROOF. Consider the probability space $\Omega = (0, 1)$, with Borel σ -field and the Uniform probability measure U of Example 1.1.11. Suffices to construct an example of $X_n \rightarrow_p 0$ such that fixing each $\omega \in (0, 1)$, we have that $X_n(\omega) = 1$ for infinitely many values of n . For example, this is the case when $X_n(\omega) = \mathbf{1}_{[t_n, t_n + s_n]}(\omega)$ with $s_n \downarrow 0$ as $n \rightarrow \infty$ slowly enough and $t_n \in [0, 1 - s_n]$ are such that any $\omega \in [0, 1]$ is in infinitely many intervals $[t_n, t_n + s_n]$. The latter property applies if $t_n = (i - 1)/k$ and $s_n = 1/k$ when $n = k(k - 1)/2 + i$, $i = 1, 2, \dots, k$ and $k = 1, 2, \dots$ (plot the intervals $[t_n, t_n + s_n]$ to convince yourself). ■

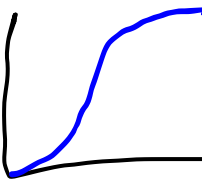
But there exists a subsequence n_k such that $X_{n_k} \xrightarrow{a.s.} X$.

Weak Convergence

Let $F_X : \mathbb{R} \rightarrow [0, 1]$ be the cumulative distribution function of real-valued random variable X .

$$F_X(x) = P(X \leq x)$$

$X_n \xrightarrow{D} X$ if $F_{X_n}(\alpha) \rightarrow F_X(\alpha)$ for each fixed α that is a continuity point of F_X .



"Weak convergence", "convergence in distribution", and "convergence in law" all mean the same thing.

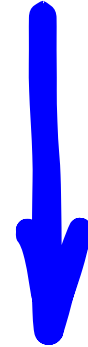
Convergence

In what senses can we talk about a sequence of random variables converging?

Stronger

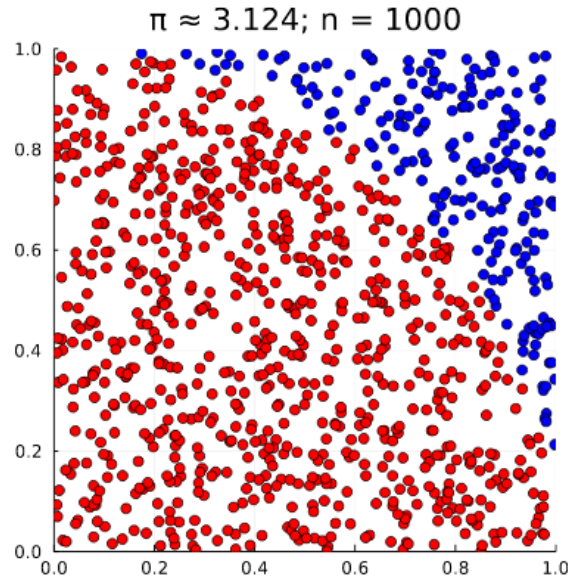


- Sure ("pointwise")
- Almost Sure
- In Probability
- Weak ("in distribution"/"in law")



Implies

Convergence of MC integration



Let X_i be independent, identically distributed random variables with mean μ , and $Q_N \equiv \frac{1}{N} \sum_{i=1}^N X_i$.

$$Q_N \xrightarrow{?} \mu?$$

Convergence of MC integration

$\exists \omega \in \Omega$ where you always sample the same point.



$$Q_N \rightarrow \mu \text{ (sure)?}$$



$$Q_N \xrightarrow{a.s.} \mu?$$



Strong law of large numbers

$$Q_N \rightarrow_p \mu?$$



Weak law of large numbers

$$Q_N \xrightarrow{D} \mu?$$



Probability that there are enough measurements off in one direction to keep $|Q_N - \mu| > \epsilon$ decays with more samples.

Convergence *Rate* of M.C. Integration

How do you quantify $|Q_N - \mu|$?

Run M sets of N simulations and plot a histogram of Q_N^j for $j \in \{1, \dots, M\}$.

Central Limit Theorem

Lindeberg-Levy CLT: If $\text{Var}[X_i] = \sigma^2 < \infty$, then

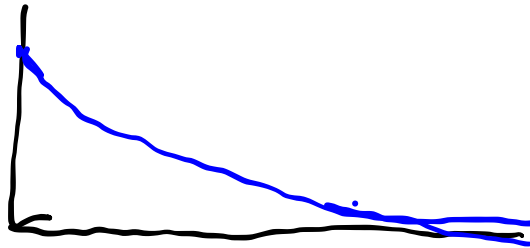
$$\sqrt{N}(Q_N - \mu) \xrightarrow{D} \mathcal{N}(0, \sigma)$$

After many samples Q_N starts to look
distributed like $\mathcal{N}(\mu, \frac{\sigma}{\sqrt{N}})$

Central Limit Theorem

Two somewhat astounding takeaways:

1. Error decays at $\frac{1}{\sqrt{N}}$ *regardless of dimension*.



2. You can estimate the "standard error" with

$$SE = \frac{s}{\sqrt{N}}$$

where s is the sample standard deviation.